TITLE: Distributed Leaderboard Management System

Aim:

To design and implement a distributed leaderboard system that maintains real-time rankings with consistent updates from multiple clients.

Abstract:

This mini project focuses on maintaining real-time rankings in a system where multiple clients can update scores simultaneously. The primary aim of this project is to design a leaderboard system that handles concurrent score updates while maintaining accurate and consistent rankings. The project is centered on network-based system design using a TCP client–server architecture.

The system is implemented using Python, allowing clients to send score updates and retrieve leaderboard data. It also addresses challenges such as concurrent updates, consistency, conflict resolution, and performance under high update rates.

Objectives:

1. To design and implement a leaderboard system using a TCP client–server architecture that allows multiple clients to send score updates and retrieve rankings.

2. To ensure accurate and consistent leaderboard rankings even when multiple clients update scores at the same time.

3. To handle concurrent updates efficiently by addressing issues such as conflict resolution and system performance under high update rates.

Methodology:

To achieve these objectives, the project utilizes Python for implementing the client-server architecture using TCP sockets. The system is designed to handle multiple client connections concurrently, ensuring synchronized updates to the leaderboard. Data was collected through simulated client requests generating score updates, and analysis was conducted based on system response time, consistency, and conflict handling efficiency.

Results:

The results of this mini project indicate that the system successfully maintains real-time leaderboard rankings with multiple concurrent updates. The implementation ensures data consistency and resolves conflicts effectively. Performance testing shows that the system can handle high update rates with minimal delay. These findings suggest that the system is suitable for applications such as gaming platforms and competitive environments.

Conclusion:

In conclusion, this mini project demonstrates the effectiveness of a distributed approach to managing real-time leaderboards. The outcomes contribute to understanding concurrent data handling and consistency in distributed systems. Future work could explore scalability improvements, advanced conflict resolution techniques, and integration with cloud-based systems for larger deployments.